

Transposed Poisson algebra structures¹

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Abstract. *Complex transposed Poisson algebra structures on some Lie and associative algebras are studied. A deep connection between unital transposed Poisson algebras and generalized Poisson algebras is established. An analog of a theorem due to Farkas for unital transposed Poisson algebras is presented. All transposed Poisson algebra structures on simple graded Lie algebras of finite growth are described. It is proved that the oscillator Lie algebras have non-trivial transposed Poisson algebra structures. Finally, a list of open questions for future research is given.*

Keywords: *Lie algebra, transposed Poisson algebra, δ -derivation.*

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INTRODUCTION

Poisson algebras arose from the study of Poisson geometry in the 1970s and have appeared in an extremely wide range of areas in mathematics and physics, such as Poisson manifolds, algebraic geometry, operads, quantization theory, quantum groups, and classical and quantum mechanics. The study of Poisson algebras also led to other algebraic structures, such as noncommutative Poisson algebras, generic Poisson algebras, algebras of Jordan brackets and generalized Poisson algebras, Gerstenhaber algebras, F -manifold algebras, Novikov-Poisson algebras, Malcev-Poisson-Jordan algebras, double Poisson algebras, Poisson n -Lie algebras, etc [11–13, 15, 26, 28, 30, 32, 37, 46, 51, 58, 63, 70]. The study of all possible Poisson structures with a certain Lie or associative part is an important problem in the theory of Poisson algebras [1, 29, 64]. Recently, a dual notion of the Poisson algebra (transposed Poisson algebra), by exchanging the roles of the two binary operations in the Leibniz rule defining the Poisson algebra, has been introduced in the paper [4] of Bai, Bai, Guo, and Wu. They have shown that the transposed Poisson algebra defined in this way not only shares common properties with the Poisson algebra, including the closure undertaking tensor products and the Koszul self-duality as an operad, but also admits a rich class of identities. More significantly, a transposed Poisson algebra naturally arises from a Novikov-Poisson algebra by taking the commutator Lie algebra of the Novikov algebra. Consequently, the classic construction of a Poisson algebra from a commutative associative algebra with a pair of commuting derivations has a similar construction to a transposed Poisson algebra when there is one derivation. More broadly, the transposed Poisson algebra also captures the algebraic structures when the commutator is taken in pre-Lie Poisson algebras and two other Poisson-type algebras. The Hom- and BiHom-versions of transposed Poisson algebras are considered in [36, 39]. Some new examples of transposed Poisson algebras are constructed by applying the Kantor product of multiplications on the same vector space [18]. More recently, in a paper by Ferreira, Kaygorodov and Lopatkin, a relation between $\frac{1}{2}$ -derivations of Lie algebras and transposed Poisson algebras has been established [20]. These ideas were used for describing all transposed Poisson structures on the Witt algebra [20], the Virasoro algebra [20], the algebra $\mathcal{W}(a, b)$ [20], the thin Lie algebra [20], the twisted Heisenberg-Virasoro algebra [65], the Schrodinger-Virasoro algebra [65], the extended Schrodinger-Virasoro algebra [65], and the 3-dimensional Heisenberg Lie algebra [65]. The complete classification of complex 3-dimensional transposed Poisson algebras is given in [7].

The class of oscillator Lie algebras, which are the only noncommutative solvable Lie algebras that carry a bi-invariant Lorentzian metric [48], is a class of quadratic Lie algebras. The former algebras, which are the Lie algebras of oscillator Lie groups, have aroused some interest [1, 5, 8–10, 14]. Ballesteros and Herranz obtained all possible coboundary Lie bialgebras for the four-dimensional oscillator algebra [5]. Biggs and Remsing examined the algebraic structure of the four-dimensional oscillator Lie algebra [8]. Firstly, the adjoint orbits are determined (and graphed); these orbits turn out to be linearly isomorphic to the coadjoint orbits. This prompts the identification of a family of invariant scalar products (from which the Casimir functions can be recovered). They identified the group of automorphisms (resp. inner automorphisms); subsequently, they classified all linear subspaces. As a by-product, they arrived at classifications of the full-rank subspaces, the subalgebras,

and the ideals. Based on the four-dimensional oscillator Lie algebra, Crampé, van de Vijver, and Vinet introduced an associative algebra which they call the oscillator Racah algebra, and explain a relation between the representation theory of and multivariate Krawtchouk polynomials [14]. In [10], Calvaruso and Zaeim obtained a classification of all Ricci, curvature, Weyl and matter collineations for the four-dimensional oscillator group equipped with a 1-parameter family of left-invariant Lorentzian metrics. Boucetta and Medina determined the Lie bialgebra structures and the solutions of the classical Yang–Baxter equation on a generic class of oscillator Lie algebras [9]. Albuquerque, Barreiro, Benayadi, Boucetta, and Sánchez described all Poisson algebra structures and symmetric Leibniz bialgebra structures on generic oscillator Lie algebras [1].

The structure of the present article is as follows. In section 1, we discuss some definitions and previous results. This section establishes many relations between transposed Poisson algebras and other well-known structures, such as generalized Poisson algebras, fields of fractions, Hom-Lie structures, quasi-automorphisms, etc. In subsection 1.2, we describe the Lie bracket in a unital transposed Poisson algebra. Namely, we prove that each bracket $[x, y]$ is given in the following way,

$$[x, y] = \mathfrak{D}(x) \cdot y - x \cdot \mathfrak{D}(y)$$

for a derivation $\mathfrak{D}$ of commutative associative part of the transposed Poisson algebra. Hence, each unital transposed Poisson algebra is a generalized Poisson algebra (and also it is an algebra of Jordan brackets). As a corollary, we have that unital transposed Poisson brackets are embeddable into Poisson brackets. At the end of this subsection, we find customary identities for unital transposed Poisson algebras (namely, we prove that each transposed Poisson PI algebra satisfies some identities of a special type). In subsection 1.3, we extend the bracket of a unital transposed Poisson algebra on their field of fractions, hence we have the notion of transposed Poisson fields. In subsection 1.4, we prove that each nilpotent finite-dimensional n -Lie algebra admits a non-trivial transposed Poisson structure. In subsection 1.5, we discuss a relation between quasi-automorphisms and transposed Poisson algebra structures. As a corollary, we have that Borel or parabolic subalgebras of a complex finite-dimensional Lie algebra of rank $l > 1$ do not admit non-trivial transposed Poisson algebra structures. In subsection 1.6, we discuss a relation between Hom-Lie structures and transposed Poisson algebra structures. As a corollary, we have that complex infinite-dimensional Lie superalgebras of vectors fields and affine Kac–Moody algebras do not admit non-trivial transposed Poisson algebra structures. In subsection 1.7, we give a complete description of transposed Poisson algebra structures on simple graded Lie algebras of finite growth. Namely, we prove that only the Witt algebra W and the Cartan algebra $W(1)$ admit non-trivial transposed Poisson algebra structures, and we describe them.

In section 2, we describe $\frac{1}{2}$ -derivations of oscillator Lie algebras. As a consequence, we have new examples of non-trivial Hom-Lie structures and transposed Poisson structures on oscillator Lie algebras. We give the full description of all non-isomorphic transposed Poisson algebra structures on generic oscillator Lie algebras.

In the ending section 3, we give a list of some open questions for future research.

1. PRELIMINARIES AND SOME TRIVIAL RESULTS

Although all algebras and vector spaces are considered over the complex field, many results can be proven over other fields without modifications of proofs. The definition of transposed Poisson algebra was given in a paper by Bai, Bai, Guo, and Wu [4]. The concept of $\frac{1}{2}$ -derivations as a particular case of δ -derivations was presented in an article of Filippov [21] (see also [27, 69] and references therein).

Definition 1. Let $\mathfrak{L}$ be a vector space equipped with two nonzero bilinear operations $\cdot$ and $[\cdot, \cdot]$. The triple $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ is called a *transposed Poisson algebra* if $(\mathfrak{L}, \cdot)$ is a commutative associative algebra and $(\mathfrak{L}, [\cdot, \cdot])$ is a Lie algebra that satisfies the following compatibility condition

$$2z \cdot [x, y] = [z \cdot x, y] + [x, z \cdot y].$$

Definition 2. Let $(\mathfrak{L}, [\cdot, \cdot])$ be an algebra with multiplication $[\cdot, \cdot]$, φ be a linear map and ϕ be a bilinear map. Then φ is a $\frac{1}{2}$ -derivation if it satisfies

$$\varphi[x, y] = \frac{1}{2}([\varphi(x), y] + [x, \varphi(y)]);$$

ϕ is a $\frac{1}{2}$ -biderivation if it satisfies

$$\begin{aligned} \phi([x, y], z) &= \frac{1}{2}([x, \phi(y, z)] + [\phi(x, z), y]), \\ \phi(x, [y, z]) &= \frac{1}{2}([\phi(x, y), z] + [\phi(x, y), z]). \end{aligned}$$

Summarizing Definitions 1 and 2, we have the following key Lemma.

Lemma 3. Let $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ be a transposed Poisson algebra and z an arbitrary element from $\mathfrak{L}$. Then the right multiplication R_z in the associative commutative algebra $(\mathfrak{L}, \cdot)$ gives a $\frac{1}{2}$ -derivation of the Lie algebra $(\mathfrak{L}, [\cdot, \cdot])$.

The main example of $\frac{1}{2}$ -derivations is the multiplication by an element from the ground field. Let us call such $\frac{1}{2}$ -derivations as trivial $\frac{1}{2}$ -derivations. As a consequence of the following theorem, we are not interested in trivial $\frac{1}{2}$ -derivations.

Theorem 4. Let $\mathfrak{L}$ be a Lie algebra without non-trivial $\frac{1}{2}$ -derivations. Then every transposed Poisson algebra structure defined on $\mathfrak{L}$ is trivial.

1.1. Isomorphism problem for transposed Poisson algebras on a certain Lie algebra.

Definition 5. Let $(\mathfrak{L}_1, \cdot_1, [\cdot, \cdot]_1)$ and $(\mathfrak{L}_2, \cdot_2, [\cdot, \cdot]_2)$ be two transposed Poisson algebras. Then $(\mathfrak{L}_1, \cdot_1, [\cdot, \cdot]_1)$ and $(\mathfrak{L}_2, \cdot_2, [\cdot, \cdot]_2)$ are isomorphic if and only if there exists a linear map φ such that

$$\varphi([x, y]_1) = [x, y]_2, \quad \varphi(x \cdot_1 y) = x \cdot_2 y.$$

Our main strategy for classifying all non-isomorphic transposed Poisson algebra structures on a certain Lie algebra $(\mathfrak{L}, [\cdot, \cdot])$ is as follows.

- (1) Find all automorphisms $\text{Aut}(\mathfrak{L}, [\cdot, \cdot])$.
- (2) Consider the multiplication table of $(\mathfrak{L}, \cdot)$ under the action of elements from $\text{Aut}(\mathfrak{L}, [\cdot, \cdot])$ and separate all non-isomorphic cases.

1.2. Transposed Poisson algebra structures on unital associative algebras. As we will see in the present subsection, unital transposed Poisson algebras are closely related to generalized Poisson algebras. Some theorems proved for generalized Poisson algebras can give some interesting results in the theory of transposed Poisson algebras.

Theorem 6. *Let $(\mathbf{A}, \cdot, [\cdot, \cdot])$ be a complex transposed Poisson algebra. Let $(\mathbf{A}, \cdot)$ be a unital associative (not necessarily commutative) algebra. Then there is a derivation $\mathfrak{D}$ on $(\mathbf{A}, \cdot)$ such that*

$$[x, y] = \mathfrak{D}(x) \cdot y - x \cdot \mathfrak{D}(y). \quad (1)$$

In particular, all transposed Poisson structures on a finite dimensional semisimple commutative associative algebra $(\mathbf{A}, \cdot)$ are trivial.

Proof. It is easy to see that

$$\begin{aligned} [x \cdot y, 1] &= 2[x, 1] \cdot y - [x, y], \\ [x \cdot y, 1] &= 2x \cdot [y, 1] - [y, x], \end{aligned}$$

then, summarizing and subtracting of it, we have

$$\begin{aligned} [x \cdot y, 1] &= x \cdot [y, 1] + [x, 1] \cdot y, \\ [x, y] &= [x, 1] \cdot y - x \cdot [y, 1]. \end{aligned}$$

Hence, the linear mapping $\mathfrak{D}$ given by $\mathfrak{D}(x) = [x, 1]$ is a derivation and the first statement of the theorem is proved.

Each finite-dimensional semisimple commutative associative algebra is a direct sum of one-dimensional ideals. On the other hand, it is known that all derivations of this algebra are trivial and it is unital. Hence, we have our statement. $\square$

Let us remember now the definition of generalized Poisson bracket and Jordan brackets [11, 28].

Definition 7. *Let $\mathbf{A}$ be a unital associative commutative algebra.*

- *An anticommutative product $\{\cdot, \cdot\} : \mathbf{A} \otimes \mathbf{A} \rightarrow \mathbf{A}$ is called a generalized Poisson bracket (also know as the contact bracket; see, [28]) if $(\mathbf{A}, \{\cdot, \cdot\})$ is a Lie algebra and*

$$\{x, y \cdot z\} = \{x, y\} \cdot z + y \cdot \{x, z\} - \{x, 1\} \cdot y \cdot z.$$

- *An anticommutative product $\{\cdot, \cdot\} : \mathbf{A} \otimes \mathbf{A} \rightarrow \mathbf{A}$ is called a Jordan bracket (see, [28]) if*

$$\{x, y \cdot z\} = \{x, y\} \cdot z + y \cdot \{x, z\} - \{x, 1\} \cdot y \cdot z;$$

$$\{a, \{b, c\}\} = \{\{a, b\}, c\} + \{b, \{a, c\}\} + \{a, 1\} \cdot \{b, c\} + \{b, 1\} \cdot \{c, a\} + \{c, 1\} \cdot \{a, b\}.$$

The main example of generalized Poisson brackets and Jordan brackets is given by relation (1). Hence, we have the following corollary.

Corollary 8. *Let $(\mathbf{A}, \cdot, [\cdot, \cdot])$ be a unital transposed Poisson algebra. Then $(\mathbf{A}, \cdot, [\cdot, \cdot])$ is a generalized Poisson algebra and an algebra of Jordan brackets.*

Martínez, Shestakov and Zel'manov have shown that all Jordan brackets are embeddable into Poisson brackets [46]. In this sense, summarizing Corollary 9 and [46], we have one more interesting statement.

Corollary 9. *Unital transposed Poisson brackets are embeddable into Poisson brackets.*

The celebrated Amitsur–Levitsky theorem states that the algebra $M_k(R)$ of $k \times k$ matrices over a commutative ring R satisfies the identity $s_{2k} = 0$. Furthermore, every associative PI algebra satisfies $(s_k)^l = 0$ by Amitsur's theorem. Farkas defined customary polynomials

$$g = \sum_{\sigma \in \mathbb{S}_m} c_\sigma \{x_{\sigma(1)}, x_{\sigma(2)}\} \cdots \{x_{\sigma(2i-1)}, x_{\sigma(2i)}\}$$

and proved that every Poisson PI algebra satisfies some customary identity [17]. Farkas' theorem was established for generic Poisson algebras in [32].

Kaygorodov found an analog of customary identities for generalized Poisson algebras and algebras of Jordan brackets [28]:

$$\mathfrak{g} = \sum_{i=0}^{\lfloor m/2 \rfloor} \sum_{\sigma \in \mathbb{S}_m} c_{\sigma,i} \langle x_{\sigma(1)}, x_{\sigma(2)} \rangle \cdots \langle x_{\sigma(2i-1)}, x_{\sigma(2i)} \rangle \mathfrak{D}(x_{\sigma(2i+1)}) \cdots \mathfrak{D}(x_{\sigma(m)}), \quad (2)$$

where

$$\langle x, y \rangle := \{x, y\} - (\mathfrak{D}(x) \cdot y - x \cdot \mathfrak{D}(y)) \text{ and } \mathfrak{D} \text{ is the related derivation, i.e., } \mathfrak{D}(x) = \{x, 1\}.$$

Theorem 10 (Theorem 13, [28]). *If a unital generalized Poisson algebra $\mathbf{A}$ satisfies a polynomial identity, then $\mathbf{A}$ satisfies a polynomial identity $\mathfrak{g}$ of type (2).*

Consequently, we obtain the following statement.

Theorem 11. *If a unital transposed Poisson algebra $\mathbf{A}$ satisfies a polynomial identity, then $\mathbf{A}$ satisfies polynomial identities $\mathfrak{g}$ and $\mathfrak{f}$ of the following types*

$$\mathfrak{g}(x_1, \dots, x_m) = \sum_{\sigma \in \mathbb{S}_m} c_\sigma \mathfrak{D}(x_{\sigma(1)}) \cdots \mathfrak{D}(x_{\sigma(m)}),$$

where $\mathfrak{D}$ is a derivation; or

$$\mathfrak{f}(x_1, \dots, x_{3m}) = \sum_{\sigma \in \mathbb{S}_m} c_\sigma \prod_{k=1}^m \left([x_{\sigma(i)} \cdot x_{m+2k-1}, x_{m+2k}] + [x_{\sigma(i)} \cdot x_{m+2k}, x_{m+2k-1}] \right).$$

Proof. Note that, in our case, $\langle x, y \rangle = 0$ and the identity (2) gives the identity $\mathfrak{g}$ from our statement.

Now, multiplying the identity (2) by $\prod_{k=m+1}^{3m} x_k$ and using the following relation, which is a consequence of the transposed Poisson identity and the generalized Poisson identity,

$$\mathfrak{D}(x) \cdot y \cdot z = -\frac{1}{2}([x \cdot y, z] + [x \cdot z, y]).$$

Hence, the identity $\mathfrak{g}$ gives the identity $\mathfrak{f}$. □

1.3. Transposed Poisson fields. Denote by $\mathcal{Q}(\mathfrak{P})$ the field of fractions of the commutative associative algebra $\mathfrak{P}$. The Poisson bracket $\{\cdot, \cdot\}$ on $\mathfrak{P}$ can be extended (see, for example, [25, 40, 44]) to a Poisson bracket on $\mathcal{Q}(\mathfrak{P})$ and

$$\left\{ \frac{a}{b}, \frac{c}{d} \right\} = \frac{\{a, c\}bd - \{a, d\}bc - \{b, c\}ad + \{b, d\}ac}{b^2d^2}.$$

Our aim in the present subsection is to define a similar notion to that of the transposed Poisson case.

Theorem 12. *Let $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ be a unital transposed Poisson algebra. Then $(\mathcal{Q}(\mathfrak{L}), \bullet, \llbracket \cdot, \cdot \rrbracket)$, where*

$$\frac{a}{b} \bullet \frac{c}{d} = \frac{a \cdot c}{b \cdot d} \quad \text{and} \quad \llbracket \frac{a}{b}, \frac{c}{d} \rrbracket = \frac{[a, b] \cdot c \cdot d - a \cdot b \cdot [c, d]}{b^2 \cdot d^2},$$

is a unital transposed Poisson field.

Proof. It is easy to see that $\bullet$ is a commutative associative multiplication and $\llbracket \cdot, \cdot \rrbracket$ is anticommutative. Hence, we should check only two identities to complete the proof (for simplification of our proof the multiplication $\cdot$ will be omitted). Thanks to theorem 6, our bracket $[x, y]$ can be written as $[x, y] = \mathfrak{D}(x)y - x\mathfrak{D}(y)$ for a derivation $\mathfrak{D}$. Hence,

$$\begin{aligned} & \frac{a}{b} \bullet \left\llbracket \frac{c}{d}, \frac{e}{f} \right\rrbracket - \frac{1}{2} \left(\left\llbracket \frac{a}{b} \bullet \frac{c}{d}, \frac{e}{f} \right\rrbracket + \left\llbracket \frac{c}{d}, \frac{a}{b} \bullet \frac{e}{f} \right\rrbracket \right) = \\ & \frac{[c, d]aef - acd[e, f]}{bd^2f^2} - \frac{[ac, bd]ef - acbd[e, f] + [c, d]aef - cd[ac, bf]}{2b^2d^2f^2} = \\ & (2b^2d^2f^2)^{-1} \left(\begin{aligned} & 2ab\mathfrak{D}(c)def - 2abc\mathfrak{D}(d)ef - 2abcd\mathfrak{D}(e)f + 2abcde\mathfrak{D}(f) - \\ & (\mathfrak{D}(a)bcdef + ab\mathfrak{D}(c)def - a\mathfrak{D}(b)cdef - abc\mathfrak{D}(d)ef) + \\ & (abcd\mathfrak{D}(e)f - abcde\mathfrak{D}(f)) - (ab\mathfrak{D}(c)def - abc\mathfrak{D}(d)ef) + \\ & (\mathfrak{D}(a)bcdef + ab\mathfrak{D}(c)def - a\mathfrak{D}(b)cdef - abcde\mathfrak{D}(f)) \end{aligned} \right) = 0. \end{aligned}$$

Now we should check the Jacobi identity. For a function $f(x_1, x_2, y_1, y_2, z_1, z_2)$, we denote

$$f(x_1, x_2, y_1, y_2, z_1, z_2) + f(y_1, y_2, z_1, z_2, x_1, x_2) + f(z_1, z_2, x_1, x_2, y_1, y_2)$$

as $\circlearrowleft f(x_1, x_2, y_1, y_2, z_1, z_2)$.

Hence,

$$\begin{aligned} & \circlearrowleft \left\llbracket \frac{x_1}{x_2}, \left\llbracket \frac{y_1}{y_2}, \frac{z_1}{z_2} \right\rrbracket \right\rrbracket = \circlearrowleft \left\llbracket \frac{x_1}{x_2}, \frac{[y_1, y_2]z_1z_2 - y_1y_2[z_1, z_2]}{y_2^2z_2^2} \right\rrbracket = \\ & \frac{\circlearrowleft ([x_1, x_2]([y_1, y_2]z_1z_2 - y_1y_2[z_1, z_2]))}{x_2^2y_2^2z_2^2} - \frac{\circlearrowleft (x_1x_2^3[[y_1, y_2]z_1z_2 - y_1y_2[z_1, z_2], y_2^2z_2^2])}{x_2^4y_2^4z_2^4}. \end{aligned}$$

We will prove that each part of the last sum is equal to zero. For the first summand of our sum, we have the following.

$$\begin{aligned} & \circ ([x_1, x_2]([y_1, y_2]z_1z_2 - y_1y_2[z_1, z_2])) = \\ & \circ (\mathfrak{D}(x_1)x_2\mathfrak{D}(y_1)y_2z_1z_2 - x_1\mathfrak{D}(x_2)\mathfrak{D}(y_1)y_2z_1z_2 + \mathfrak{D}(x_1)x_2y_1\mathfrak{D}(y_2)z_1z_2 - \\ & \quad x_1\mathfrak{D}(x_2)y_1\mathfrak{D}(y_2)z_1z_2 - \mathfrak{D}(x_1)x_2y_1y_2\mathfrak{D}(z_1)z_2 + \mathfrak{D}(x_1)x_2y_1y_2z_1\mathfrak{D}(z_2) - \\ & \quad x_1\mathfrak{D}(x_2)y_1y_2\mathfrak{D}(z_1)z_2 + x_1\mathfrak{D}(x_2)y_1y_2z_1\mathfrak{D}(z_2)) = 0. \end{aligned}$$

We will rewrite the second summand in a sum of two other sums in the following way.

$$\begin{aligned} & \circ (x_1x_2^3([y_1, y_2]z_1z_2 - y_1y_2[z_1, z_2], y_2^2z_2^2)) = \\ & \circ (x_1x_2^3(\mathfrak{D}(y_1)y_2z_1z_2 - y_1\mathfrak{D}(y_2)z_1z_2 - y_1y_2\mathfrak{D}(z_1)z_2 + y_1y_2z_1\mathfrak{D}(z_2), y_2^2z_2^2)) = \\ & \quad \circ x_1x_2^3(\mathfrak{D}(\mathfrak{D}(y_1)y_2z_1z_2)y_2^2z_2^2 - \mathfrak{D}(y_1)y_2z_1z_2\mathfrak{D}(y_2^2z_2^2) - \\ & \quad \mathfrak{D}(y_1\mathfrak{D}(y_2)z_1z_2)y_2^2z_2^2 + y_1\mathfrak{D}(y_2)z_1z_2\mathfrak{D}(y_2^2z_2^2) - \\ & \quad \mathfrak{D}(y_1y_2\mathfrak{D}(z_1)z_2)y_2^2z_2^2 + y_1y_2\mathfrak{D}(z_1)z_2\mathfrak{D}(y_2^2z_2^2) + \\ & \quad \mathfrak{D}(y_1y_2z_1\mathfrak{D}(z_2))y_2^2z_2^2 - y_1y_2z_1\mathfrak{D}(z_2)\mathfrak{D}(y_2^2z_2^2)). \end{aligned}$$

Hence, the first subsum is given by the following terms.

$$\begin{aligned} & \circ x_1x_2^3y_2^2z_2^2(\mathfrak{D}(\mathfrak{D}(y_1)y_2z_1z_2) - \mathfrak{D}(y_1\mathfrak{D}(y_2)z_1z_2) - \mathfrak{D}(y_1y_2\mathfrak{D}(z_1)z_2) + \mathfrak{D}(y_1y_2z_1\mathfrak{D}(z_2))) = \\ & \circ x_1x_2^3y_2^2z_2^2(\mathfrak{D}^2(y_1)y_2z_1z_2 + \mathfrak{D}(y_1)\mathfrak{D}(y_2)z_1z_2 + \mathfrak{D}(y_1)y_2\mathfrak{D}(z_1)z_2 + \mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) - \\ & \quad \mathfrak{D}(y_1)\mathfrak{D}(y_2)z_1z_2 - y_1\mathfrak{D}^2(y_2)z_1z_2 - y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2 - y_1\mathfrak{D}(y_2)z_1\mathfrak{D}(z_2) - \\ & \quad \mathfrak{D}(y_1)y_2\mathfrak{D}(z_1)z_2 - y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2 - y_1y_2\mathfrak{D}^2(z_1)z_2 - y_1y_2\mathfrak{D}(z_1)\mathfrak{D}(z_2) + \\ & \quad \mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) + y_1\mathfrak{D}(y_2)z_1\mathfrak{D}(z_2) + y_1y_2\mathfrak{D}(z_1)\mathfrak{D}(z_2) + y_1y_2z_1\mathfrak{D}^2(z_2)) = \\ & \quad \circ (x_1\mathfrak{D}^2(y_1)z_1 - x_1y_1\mathfrak{D}^2(z_1))(x_2y_2z_2)^3 + \\ & \quad \circ (x_2\mathfrak{D}^2(y_2)z_2 - x_2y_2\mathfrak{D}^2(z_2))x_1y_1z_1(x_2y_2z_2)^2 + \\ & \quad 2 \circ (x_1x_2\mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) - x_1x_2y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2)(x_2y_2z_2)^3 = \\ & \quad 2 \circ (x_1x_2\mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) - x_1x_2y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2)(x_2y_2z_2)^2. \end{aligned}$$

And the second subsum is given by the following terms.

$$\begin{aligned} & \circ x_1x_2^3\mathfrak{D}(y_2^2z_2^2)(-\mathfrak{D}(y_1)y_2z_1z_2 + y_1\mathfrak{D}(y_2)z_1z_2 + y_1y_2\mathfrak{D}(z_1)z_2 - y_1y_2z_1\mathfrak{D}(z_2)) = \\ & \quad 2 \circ x_1x_2^3(-\mathfrak{D}(y_1)y_2^2\mathfrak{D}(y_2)z_1z_2^3 - \mathfrak{D}(y_1)y_2^3z_1z_2^2\mathfrak{D}(z_2) + y_1y_2(\mathfrak{D}(y_2))^2z_1z_2^3 + \\ & \quad y_1y_2^2\mathfrak{D}(y_2)z_1z_2^2\mathfrak{D}(z_2) + y_1y_2^2\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2^3 + y_1y_2^3\mathfrak{D}(z_1)z_2^2\mathfrak{D}(z_2) - \\ & \quad y_1y_2^2\mathfrak{D}(y_2)z_1z_2^2\mathfrak{D}(z_2) - y_1y_2^3z_1z_2^2(\mathfrak{D}(z_2))^2) = \\ & \quad 2 \circ (x_2^2(\mathfrak{D}(y_2))^2z_2^2 - x_2^2y_2^2(\mathfrak{D}(z_2))^2)x_1x_2y_1y_2z_1z_2 - \\ & \quad 2 \circ (x_1x_2\mathfrak{D}(y_1)\mathfrak{D}(y_2)z_1z_2 - x_1x_2y_1y_2\mathfrak{D}(z_1)\mathfrak{D}(z_2))(x_2y_2z_2)^2 - \\ & \quad 2 \circ (x_1x_2\mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) - x_1x_2y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2)(x_2y_2z_2)^2 = \\ & \quad -2 \circ (x_1x_2\mathfrak{D}(y_1)y_2z_1\mathfrak{D}(z_2) - x_1x_2y_1\mathfrak{D}(y_2)\mathfrak{D}(z_1)z_2)(x_2y_2z_2)^2. \end{aligned}$$

To summarize, the sum of these two subsums gives zero and we have that the Jacobi identity holds and we finish the proof. $\square$

1.4. Transposed Poisson structures on nilpotent n -Lie algebras. Let us recall the definition of transposed Poisson structures on n -Lie algebras.

Definition 13. Let $\mathfrak{L}$ be a vector space equipped with two nonzero operations – a bilinear operation $\cdot$ and an n -linear operation $[\cdot, \dots, \cdot]$. The triple $(\mathfrak{L}, \cdot, [\cdot, \dots, \cdot])$ is called a transposed Poisson n -Lie algebra if $(\mathfrak{L}, \cdot)$ is a commutative associative algebra and $(\mathfrak{L}, [\cdot, \dots, \cdot])$ is an n -Lie algebra that satisfies the following compatibility condition

$$z \cdot [x_1, \dots, x_n] = \frac{1}{n} \sum_{i=1}^n [x_1, \dots, z \cdot x_i, \dots, x_n].$$

Theorem 14. Let $\mathfrak{L}$ be a nilpotent k -dimensional n -Lie algebra. Then $\mathfrak{L}$ admits a non-trivial transposed Poisson n -Lie structure.

Proof. It is easy to see that, in the case $n \geq k$, $\mathfrak{L}$ has the zero multiplication and, obviously, it admits a non-trivial transposed Poisson n -Lie structure. Let us now consider only the case $n < k$. Choose the basis $\{e_1, e_2, \dots, e_k\}$ of $\mathfrak{L}$ in the following way: $e_1, e_2, \dots, e_n \in \mathfrak{L} \setminus \mathfrak{L}^2$ and $e_k \in \text{Ann}(\mathfrak{L})$. It is easy to see that the linear mapping φ given by $\varphi(e_1) = \varphi(e_2) = \dots = \varphi(e_n) = e_k$ is a $\frac{1}{n}$ -derivation of $\mathfrak{L}$. Then the commutative associative multiplication $\cdot$ given by $e_i \cdot e_j = e_k$ ($1 \leq i, j \leq n$) gives a non-trivial structure of transposed Poisson n -Lie algebra. $\square$

1.5. Transposed Poisson structures and quasi-automorphisms. Let us recall the definition of quasi-automorphisms on Lie algebras.

Definition 15. Let $(\mathfrak{L}, [\cdot, \cdot])$ be a Lie algebra and φ be a linear map. Then φ is a quasi-automorphism on $(\mathfrak{L}, [\cdot, \cdot])$ if there exist another linear map ϕ , such that

$$[\varphi(x), \varphi(y)] = \phi[x, y].$$

Thanks to [4, Proposition 2.9], each transposed Poisson structure $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ introduces a quasi-automorphism φ_h on $(\mathfrak{L}, [\cdot, \cdot])$, such that

$$\varphi_h^2[x, y] = [\varphi_h(x), \varphi_h(y)], \text{ where } \varphi_h(x) = h \cdot x \text{ for a fixed element } h.$$

Automorphisms and scalar multiplication maps are trivial quasi-automorphisms. Hence, we have the following statement.

Proposition 16. If a Lie algebra $(\mathfrak{L}, [\cdot, \cdot])$ does not admit non-trivial quasi-automorphisms, then $(\mathfrak{L}, [\cdot, \cdot])$ does not admit non-trivial transposed Poisson structures.

Zhou, Feng, Kong, Wang and Chen gave the classification of quasi-automorphisms of Borel and parabolic subalgebras of a complex finite-dimensional simple Lie algebra of rank l . Namely, they proved that in the case $l > 1$, each quasi-automorphism is trivial [60, 68]. Summarizing with Proposition 16, we have the following statement.

Lemma 17. Let $\mathfrak{L}$ be a complex finite-dimensional Lie algebra of rank $l > 1$ and $\mathfrak{p}$ be an arbitrary Borel or parabolic subalgebra of $\mathfrak{L}$. Then $\mathfrak{p}$ does not admit non-trivial transposed Poisson algebra structures.

1.6. Transposed Poisson structures and Hom-Lie algebra structures. Let us recall the definition of Hom-structures on Lie algebras.

Definition 18. Let $(\mathfrak{L}, [\cdot, \cdot])$ be a Lie algebra and φ be a linear map. Then $(\mathfrak{L}, [\cdot, \cdot], \varphi)$ is a Hom-Lie structure on $(\mathfrak{L}, [\cdot, \cdot])$ if

$$[\varphi(x), [y, z]] + [\varphi(y), [z, x]] + [\varphi(z), [x, y]] = 0.$$

Let us give some special cases of Hom-Lie structures.

- $(\mathfrak{L}, [\cdot, \cdot], \varphi)$ is a trivial Hom-Lie structure on $(\mathfrak{L}, [\cdot, \cdot])$ if $\varphi = \mathbb{C} \cdot \text{id}$.
- $(\mathfrak{L}, [\cdot, \cdot], \varphi)$ is a central Hom-Lie structure on $(\mathfrak{L}, [\cdot, \cdot])$ if $\varphi(\mathfrak{L}) \subseteq Z(\mathfrak{L})$.
- $(\mathfrak{L}, [\cdot, \cdot], \varphi)$ is a $\mathfrak{c}$ -trivial Hom-Lie structure on $(\mathfrak{L}, [\cdot, \cdot])$ if φ is a linear combination of trivial and central Hom-Lie structures.

Thanks to [4, Proposition 2.9], each transposed Poisson structure $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ introduces a Hom-Lie structure φ_h on $(\mathfrak{L}, [\cdot, \cdot])$, such that $\varphi_h(x) = h \cdot x$ for a fixed element h . Hence, we have the following statement.

Proposition 19. If a Lie algebra $(\mathfrak{L}, [\cdot, \cdot])$ does not admit non-trivial Hom-Lie structures, then $(\mathfrak{L}, [\cdot, \cdot])$ does not admit non-trivial transposed Poisson structures.

The subsequent list of algebras gives algebras without non-trivial Hom-Lie structures (known until now):

- (1) complex finite-dimensional simple Lie algebras ($\dim \neq 3$) and superalgebras [24, 61, 66];
- (2) infinite-dimensional Lie superalgebras of vectors fields [54, 67];
- (3) infinite-dimensional Cartan algebras $W(n > 1)$, $S(n)$, $H(2r)$, $K(2r + 1)$ [62].

Hence, using Proposition 19, we have the following statement.

Lemma 20. Complex finite-dimensional simple Lie algebras ($\dim \neq 3$) and superalgebras, infinite-dimensional Lie superalgebras of vectors fields, and infinite-dimensional Cartan algebras $W(n > 1)$, $S(n)$, $H(2r)$, $K(2r + 1)$ do not admit non-trivial transposed Poisson structures.

Let us give one more useful trivial observation.

Proposition 21. If a Lie algebra $(\mathfrak{L}, [\cdot, \cdot])$ with one-dimensional center $Z(\mathfrak{L})$ admits only $\mathfrak{c}$ -trivial Hom-Lie structures, then $(\mathfrak{L}, [\cdot, \cdot])$ does not admit non-trivial transposed Poisson structures.

Thanks to [45], all Hom-Lie structures on affine Kac–Moody algebras are $\mathfrak{c}$ -trivial. Hence, summarizing Proposition 21 and results from [45], we have the following statement.

Lemma 22. Complex affine Kac–Moody algebras do not admit non-trivial transposed Poisson algebra structures.

1.7. Transposed Poisson structures on simple graded Lie algebras of finite growth. Let us recall the definition of a complex simple graded Lie algebra of finite growth. A complex graded Lie algebra is a Lie algebra $\mathfrak{L}$ endowed with a decomposition $\mathfrak{L} = \bigoplus \mathfrak{L}_i$, such that $[\mathfrak{L}_i, \mathfrak{L}_j] \subseteq \mathfrak{L}_{i+j}$ for every $i, j \in \mathbb{Z}$. We always assume that every homogeneous component $\mathfrak{L}_i$ is of finite dimension. A graded Lie algebra $\mathfrak{L}$ is called simple graded if $\mathfrak{L}$ is not abelian and does not contain any non-trivial graded ideal. We say that $\mathfrak{L}$ is of finite growth if the function $n \rightarrow \dim \mathfrak{L}_n$ is bounded by some polynomial. In [47], Mathieu proved the following theorem, which was conjectured by Kac.

Theorem 23. *Let $\mathfrak{L}$ be a graded simple Lie algebra of finite growth. Then $\mathfrak{L}$ is isomorphic to one of the following Lie algebras:*

- (1) *A finite-dimensional simple Lie algebra;*
- (2) *A loop algebra;*
- (3) *A Cartan algebra;*
- (4) *The Witt algebra W .*

Xie and Liu described all Hom-Lie structures on simple graded Lie algebras of finite growth [62]. Summarizing results [47] and [62], it is possible to prove the following theorem.

Theorem 24. *Let $\mathfrak{L}$ be a complex graded simple Lie algebra of finite growth. If $\mathfrak{L}$ admits a non-trivial transposed Poisson algebra structure, then it is isomorphic to one of the following Lie algebras:*

- (1) *The Witt algebra W , or*
- (2) *The Cartan algebra $W(1)$.*

Proof. Thanks to [62], we know that only a loop algebra, the Witt algebra W and the Cartan algebra $W(1)$ have non-trivial Hom-Lie structures.

- It is easy to see (from the description of Hom-Lie structures), that a loop algebra does not admit non-trivial transposed Poisson algebra structures.
- All non-trivial transposed Poisson algebra structures on the Witt algebra are described in [20, Theorem 20].
- The Cartan algebra $W(1)$ admits the following multiplication table

$$[e_i, e_j] = (i - j)e_{i+j}, \quad i, j \geq -1.$$

Using the same idea that was used for the description of transposed Poisson algebra structures on the Witt algebra [20, Theorem 20], we have that each transposed Poisson structure on $W(1)$ is given by the following additional multiplication

$$e_i \cdot e_j = \sum_{t \geq 0} \alpha_t e_{i+j+t},$$

for a finite set $\{\alpha_t\}$. On the other hand, each finite set $\{\alpha_t\}$ provides a structure of transposed Poisson algebra by the given way.

□

Remark 25. *Let us note that the Cartan algebra $W(1)$ gives an example of a simple Lie algebra that admits only non-unital transposed Poisson structures.*

2. $\frac{1}{2}$ -DERIVATIONS OF OSCILLATOR LIE ALGEBRAS AND TRANSPOSED POISSON ALGEBRAS

Definition 26. The oscillator Lie algebra $\mathfrak{L}_\lambda$ is a $(2n+2)$ -dimensional Lie algebra with its canonical basis $\mathbb{B} = \{e_{-1}, e_0, e_j, \check{e}_j\}_{j=1, \dots, n}$ and the Lie brackets are given by

$$[e_{-1}, e_j] = \lambda_j \check{e}_j, \quad [e_{-1}, \check{e}_j] = -\lambda_j e_j, \quad [e_j, \check{e}_j] = e_0,$$

for $j = 1, \dots, n$ and $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$ with $0 < \lambda_1 \leq \dots \leq \lambda_n$.

Proposition 27. A linear map $\varphi : \mathfrak{L}_\lambda \rightarrow \mathfrak{L}_\lambda$ is a $\frac{1}{2}$ -derivation of the algebra $(\mathfrak{L}_\lambda, [\cdot, \cdot])$ if and only if

$$\begin{aligned} \varphi(e_{-1}) &= \gamma e_{-1} + \mu e_0 - \sum_{j=1}^n 2\lambda_j \alpha_j e_j - \sum_{j=1}^n 2\lambda_j \beta_j \check{e}_j, \\ \varphi(e_0) &= \gamma e_0, \\ \varphi(e_j) &= \alpha_j e_0 + \gamma e_j, \quad (j = 1, \dots, n), \\ \varphi(\check{e}_j) &= \beta_j e_0 + \gamma \check{e}_j, \quad (j = 1, \dots, n). \end{aligned}$$

Proof. For each $\ell \in \mathbb{B}$, we consider

$$\varphi(\ell) = \varphi(\ell)_{-1} e_{-1} + \varphi(\ell)_0 e_0 + \sum_{k=1}^n (\varphi(\ell)_k e_k + \varphi(\ell)_{\check{k}} \check{e}_k).$$

Since $e_0 = [e_j, \check{e}_j]$ for each $j \in \{1, \dots, n\}$, we get

$$\begin{aligned} \varphi(e_0) &= \varphi([e_j, \check{e}_j]) = \frac{1}{2}([\varphi(e_j), \check{e}_j] + [e_j, \varphi(\check{e}_j)]) \\ &= \frac{1}{2}(-\lambda_j \varphi(e_j)_{-1} e_j + \varphi(e_j)_j e_0 - \lambda_j \varphi(\check{e}_j)_{-1} \check{e}_j + \varphi(\check{e}_j)_{\check{j}} e_0). \end{aligned}$$

Thus, $\varphi(e_0) = \gamma e_0$, where $\gamma = \frac{1}{2}(\varphi(e_j)_j + \varphi(\check{e}_j)_{\check{j}})$.

It is easy to see that

$$\begin{aligned} \lambda_j^2 \varphi(e_j) &= \varphi([e_{-1}, [e_j, e_{-1}]]) = -\frac{1}{2} \lambda_j [\varphi(e_{-1}), \check{e}_j] - \frac{1}{4} [e_{-1}, [e_{-1}, \varphi(e_j)]] + \frac{1}{4} [e_{-1}, [e_j, \varphi(e_{-1})]] \\ &= \frac{1}{2} \lambda_j^2 \varphi(e_{-1})_{-1} e_j - \frac{1}{2} \lambda_j \varphi(e_{-1})_j e_0 + \frac{1}{4} \sum_{t=1}^n \lambda_t^2 (\varphi(e_j)_t + \varphi(e_j)_{\check{t}} \check{e}_t) + \frac{1}{4} \lambda_j^2 \varphi(e_{-1})_{-1} e_j, \end{aligned}$$

hence,

$$\varphi(e_{-1})_{-1} = \varphi(e_j)_j, \quad \varphi(e_{-1})_j = -2\lambda_j \varphi(e_j)_0 \text{ and } \varphi(e_j)_t = 0, \text{ for } t \in \{-1, 1, \dots, n, \check{1}, \dots, \check{n}\} \setminus \{j\}.$$

In a similar way, we have

$$\varphi(e_{-1})_{-1} = \varphi(\check{e}_j)_{\check{j}}, \quad \varphi(e_{-1})_{\check{j}} = -2\lambda_j \varphi(\check{e}_j)_0 \text{ and } \varphi(\check{e}_j)_t = 0, \text{ for } t \in \{-1, 1, \dots, n, \check{1}, \dots, \check{n}\} \setminus \{\check{j}\}.$$

Hence, our map φ has the following type:

$$\begin{aligned} \varphi(e_{-1}) &= \gamma e_{-1} + \mu e_0 - \sum_{j=1}^n 2\lambda_j \alpha_j e_j - \sum_{j=1}^n 2\lambda_j \beta_j \check{e}_j, \\ \varphi(e_0) &= \gamma e_0, \\ \varphi(e_j) &= \alpha_j e_0 + \gamma e_j, \quad (j = 1, \dots, n), \\ \varphi(\check{e}_j) &= \beta_j e_0 + \gamma \check{e}_j, \quad (j = 1, \dots, n). \end{aligned}$$

It is easy to see that all linear maps given by the present way are $\frac{1}{2}$ -derivations. It finishes the proof. $\square$

Filippov proved that each δ -derivation ($\delta \neq 0, 1$) gives a non-trivial Hom-Lie algebra structure [21, Theorem 1]. Hence, by Proposition 27, we have the following corollary.

Corollary 28. *The oscillator Lie algebra $\mathfrak{L}_\lambda$ admits non-trivial Hom-Lie algebra structures.*

Theorem 29. *Let $(\mathfrak{L}_\lambda, [\cdot, \cdot])$ be the oscillator Lie algebra. Then every symmetric $\frac{1}{2}$ -biderivation gives a transposed Poisson algebra structure. If $(\mathfrak{L}_\lambda, \cdot, [\cdot, \cdot])$ is a transposed Poisson algebra, then $(\mathfrak{L}_\lambda, \cdot)$ has the following multiplication:*

$$\begin{aligned} e_{-1} \cdot e_{-1} &= \gamma e_{-1} + \mu e_0 - 2 \sum_{k=1}^n \lambda_k (\alpha_k e_k + \beta_k \check{e}_k), \\ e_{-1} \cdot e_0 &= \gamma e_0, \\ e_{-1} \cdot e_j &= \alpha_j e_0 + \gamma e_j, \\ e_{-1} \cdot \check{e}_j &= \beta_j e_0 + \gamma \check{e}_j, \\ e_j \cdot e_j &= \check{e}_j \cdot \check{e}_j = -\frac{\gamma}{2\lambda_j} e_0. \end{aligned}$$

$(\mathfrak{L}_\lambda, \cdot, [\cdot, \cdot])$ is a Poisson algebra if and only if $(\gamma, \alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n) = 0$.

Proof. We aim to describe the multiplication $\cdot$. By Lemma 3, for each $\ell \in \mathbb{B}$, there is a related $\frac{1}{2}$ -derivation φ_ℓ of $(\mathfrak{L}_\lambda, [\cdot, \cdot])$ such that $\varphi_{\ell_1}(\ell_2) = \ell_1 \cdot \ell_2 = \varphi_{\ell_2}(\ell_1)$, with $\ell_1, \ell_2 \in \mathbb{B}$. By Proposition 27, for each $\ell \in \mathbb{B}$, we have

$$\begin{aligned} \varphi_\ell(e_{-1}) &= \gamma^\ell e_{-1} + \mu^\ell e_0 - \sum_{k=1}^n 2\lambda_k \alpha_k^\ell e_k - \sum_{k=1}^n 2\lambda_k \beta_k^\ell \check{e}_k, \\ \varphi_\ell(e_0) &= \gamma^\ell e_0, \\ \varphi_\ell(e_j) &= \alpha_j^\ell e_0 + \gamma^\ell e_j, \quad (j = 1, \dots, n), \\ \varphi_\ell(\check{e}_j) &= \beta_j^\ell e_0 + \gamma^\ell \check{e}_j, \quad (j = 1, \dots, n). \end{aligned}$$

In the first, for all $i, j \in \{1, \dots, n\}$, we have

$$\begin{aligned} \alpha_j^{e_i} e_0 + \gamma^{e_i} e_j &= \varphi_{e_i}(e_j) = \varphi_{e_j}(e_i) = \alpha_i^{e_j} e_0 + \gamma^{e_j} e_i, \\ \alpha_j^{\check{e}_i} e_0 + \gamma^{\check{e}_i} e_j &= \varphi_{\check{e}_i}(e_j) = \varphi_{e_j}(\check{e}_i) = \beta_i^{e_j} e_0 + \gamma^{e_j} \check{e}_i, \\ \beta_j^{\check{e}_i} e_0 + \gamma^{\check{e}_i} \check{e}_j &= \varphi_{\check{e}_i}(\check{e}_j) = \varphi_{\check{e}_j}(\check{e}_i) = \beta_i^{\check{e}_j} e_0 + \gamma^{\check{e}_j} \check{e}_i, \end{aligned}$$

hence $\beta_i^{\check{e}_j} = \beta_j^{\check{e}_i}$, $\alpha_j^{e_i} = \alpha_i^{e_j}$, $\alpha_j^{\check{e}_i} = \beta_i^{e_j}$ and $\gamma^{e_j} = \gamma^{\check{e}_j} = 0$.

Second, for all $j \in \{1, \dots, n\}$, we have

$$\begin{aligned} \gamma^{e_{-1}} e_0 &= \varphi_{e_{-1}}(e_0) = \varphi_{e_0}(e_{-1}) = \gamma^{e_0} e_{-1} + \mu^{e_0} e_0 - 2 \sum_{k=1}^n \lambda_k (\alpha_k^{e_0} e_k + \beta_k^{e_0} \check{e}_k), \\ 0 &= \varphi_{e_j}(e_0) = \varphi_{e_0}(e_j) = \alpha_j^{e_0} e_0 + \gamma^{e_0} e_j, \\ 0 &= \varphi_{\check{e}_j}(e_0) = \varphi_{e_0}(\check{e}_j) = \beta_j^{e_0} e_0 + \gamma^{e_0} \check{e}_j, \end{aligned}$$

hence $\alpha_j^{e_0} = \beta_j^{e_0} = \gamma^{e_0} = 0$ and $\gamma^{e_{-1}} = \mu^{e_0}$.

Also, for all $j \in \{1, \dots, n\}$, we have

$$\begin{aligned}\alpha_j^{e-1}e_0 + \gamma^{e-1}e_j &= \varphi_{e-1}(e_j) = \varphi_{e_j}(e_{-1}) = \mu^{e_j}e_0 - 2\sum_{k=1}^n \lambda_k(\alpha_k^{e_j}e_k + \beta_k^{e_j}\check{e}_k), \\ \beta_j^{e-1}e_0 + \gamma^{e-1}\check{e}_j &= \varphi_{e-1}(\check{e}_j) = \varphi_{\check{e}_j}(e_{-1}) = \mu^{\check{e}_j}e_0 - 2\sum_{k=1}^n \lambda_k(\alpha_k^{\check{e}_j}e_k + \beta_k^{\check{e}_j}\check{e}_k),\end{aligned}$$

hence

$$\begin{aligned}\gamma^{e-1} &= -2\lambda_j\alpha_j^{e_j}, \quad \alpha_j^{e_j} = \beta_j^{\check{e}_j}, \quad \alpha_k^{\check{e}_j} = 0, \quad \beta_k^{e_j} = 0, \\ \alpha_k^{e_j} &= 0 \ (j \neq k), \quad \beta_k^{\check{e}_j} = 0 \ (j \neq k), \quad \alpha_j^{e-1} = \mu^{e_j}, \quad \beta_j^{e-1} = \mu^{\check{e}_j}.\end{aligned}$$

It follows that, for all $j \in \{1, \dots, n\}$, the multiplication $\cdot$ can be written as

$$\begin{aligned}e_{-1} \cdot e_{-1} &= \gamma e_{-1} + \mu e_0 - 2\sum_{k=1}^n \lambda_k(\alpha_k e_k + \beta_k \check{e}_k), \\ e_{-1} \cdot e_0 &= \gamma e_0, \\ e_{-1} \cdot e_j &= \alpha_j e_0 + \gamma e_j, \\ e_{-1} \cdot \check{e}_j &= \beta_j e_0 + \gamma \check{e}_j, \\ e_j \cdot e_j &= \check{e}_j \cdot \check{e}_j = -\frac{\gamma}{2\lambda_j} e_0.\end{aligned}$$

It is easy to see that $\cdot$ is a commutative associative multiplication. □

2.1. Isomorphism problem for transposed Poisson algebras on generic oscillator Lie algebras.

An oscillator Lie algebra is called generic if $0 < \lambda_1 < \dots < \lambda_n$ and $\lambda_i + \lambda_j \neq \lambda_k$ for all $1 \leq i < j < k \leq n$. It is easy to see that $(\mathfrak{L}_{\{\lambda_1, \lambda_2, \dots, \lambda_n\}}, [\cdot, \cdot])$ is isomorphic to $(\mathfrak{L}_{\{1, \frac{\lambda_2}{\lambda_1}, \dots, \frac{\lambda_n}{\lambda_1}\}}, [\cdot, \cdot])$. Hence, we will consider only the case $\lambda_1 = 1$.

For proving the main result of this part of the paper, we should give the following useful Lemma.

Lemma 30. *Let ϕ be an automorphism of $(\mathfrak{L}_\lambda, [\cdot, \cdot])$. Then ϕ has the following types:*

$$\begin{aligned}\phi(e_{-1}) &= \pm e_{-1} + \nu e_0 + \sum_{i=1}^n \nu_i e_i + \sum_{i=1}^n \check{\nu}_i \check{e}_i, \\ \phi(e_0) &= \pm \xi e_0, \\ \phi(e_i) &= \frac{\check{\nu}_i \mu_i \mp \nu_i \mu_i}{\lambda_i} e_0 + \mu_i e_i \mp \check{\mu}_i \check{e}_i, \\ \phi(\check{e}_i) &= \frac{-\check{\nu}_i \mu_i \mp \nu_i \mu_i}{\lambda_i} e_0 + \check{\mu}_i e_i \pm \mu_i \check{e}_i,\end{aligned}$$

where $\xi = \mu_i^2 + \check{\mu}_i^2$.

Proof. Let us say that

$$\phi(e_m) = \alpha_{-1}^m e_{-1} + \alpha_0^m e_0 + \sum_{i=1}^n (\alpha_i^m e_i + \beta_i^m \check{e}_i), \text{ where } m \in \{-1, 0, 1, \dots, n\}.$$

Then, for each $j \in \{1, \dots, n\}$, we have

$$\lambda_j^2 \phi(e_j) = -[\phi(e_{-1}), [\phi(e_{-1}), \phi(e_j)]] = (\alpha_{-1}^{-1})^2 \sum_{i=1}^n \lambda_i^2 (\alpha_i^j e_i + \beta_i^j \check{e}_j) + \alpha_{-1}^{-1} \sum_{i=1}^n (\alpha_i^{-1} \alpha_i^j - \beta_i^{-1} \beta_i^j) e_0.$$

Hence $\lambda_i \neq \lambda_j$, we have that $\phi(e_j) = \alpha_0^j e_0 + \alpha_j^j e_j + \beta_j^j \check{e}_j$ and $\alpha_{-1}^{-1} = \pm 1$. In a similar way, it is possible to find that $\phi(\check{e}_j)$ has a same type. Obvioulsy, ϕ has the form from our statement. $\square$

For our main theorem from this part, we need to define the following set

$$\mathbb{C}_{>0} = \{a + bi\}_{a>0} \cup \{bi\}_{b \geq 0}.$$

Theorem 31. *Let $(\mathfrak{L}_\lambda, [\cdot, \cdot])$ be the generic oscillator Lie algebra. If $(\mathfrak{L}_\lambda, \cdot, [\cdot, \cdot])$ is a transposed Poisson algebra, then $(\mathfrak{L}_\lambda, \cdot)$ has the following multiplication.*

(A) *If $(\mathfrak{L}_\lambda, \cdot)$ is non-nilpotent, then, for a $\gamma \neq 0$, we have:*

$$e_{-1} \cdot e_{-1} = \gamma e_{-1}, \quad e_{-1} \cdot e_0 = \gamma e_0, \quad e_{-1} \cdot e_j = \gamma e_j, \quad e_{-1} \cdot \check{e}_j = \gamma \check{e}_j, \quad e_j \cdot e_j = \check{e}_j \cdot \check{e}_j = -\frac{\gamma}{2\lambda_j} e_0,$$

corresponding transposed Poisson algebra $(\mathfrak{L}_{\{\lambda, \gamma\}}, \cdot, [\cdot, \cdot])$ is isomorphic only to $(\mathfrak{L}_{\{\lambda, -\gamma\}}, \cdot, [\cdot, \cdot])$; all these algebras are non-Poisson.

(B) *If $(\mathfrak{L}_\lambda, \cdot)$ is nilpotent, then there is a sequence $\{\beta_q \in \mathbb{C}_{>0}\}_{1 \leq q \leq n}$, where the first nonzero element is equal to 1, and $(\mathfrak{L}_\lambda, \cdot)$ has the following multiplication table:*

$$(a) \quad e_{-1} \cdot e_{-1} = e_0 - 2 \sum_{j=1}^n \lambda_j \beta_j \check{e}_j, \quad e_{-1} \cdot \check{e}_j = \beta_j \check{e}_j,$$

$$(b) \quad e_{-1} \cdot e_{-1} = -2 \sum_{j=1}^n \lambda_j \beta_j \check{e}_j, \quad e_{-1} \cdot \check{e}_j = \beta_j \check{e}_j.$$

The corresponding transposed Poisson algebras $(\mathfrak{L}_{\lambda, \beta}, \cdot, [\cdot, \cdot])$ are not isomorphic; and they are non-Poisson if there exists $\beta_j \neq 0$.

Proof. Let us consider the transposed Poisson structure $(\mathfrak{L}_\lambda, \cdot, [\cdot, \cdot])$ on the generic oscillator Lie algebra $(\mathfrak{L}_\lambda, [\cdot, \cdot])$ and $\cdot$ is given as in theorem 29. Under the action of a positive automorphism of the Lie algebra $(\mathfrak{L}_\lambda, [\cdot, \cdot])$, given in Lemma 30, we rewrite the multiplication table of $(\mathfrak{L}_\lambda, \cdot)$ by the following way:

$$\begin{aligned} e_{-1} \cdot e_{-1} &= \gamma e_{-1} + \left(\frac{\mu + \gamma \nu}{\xi} + \sum_{i=1}^n \frac{\gamma(\nu_i^2 + \check{\nu}_i^2)}{2\lambda_i \xi} \right) e_0 + \\ &\quad \sum_{k=1}^n \left(\frac{\gamma(\mu_k \nu_k - \check{\mu}_k \check{\nu}_k) - 2\lambda_k(\alpha_k \mu_k - \beta_k \check{\mu}_k)}{\xi} \right) e_k + \sum_{k=1}^n \left(\frac{\gamma(\check{\mu}_k \nu_k + \mu_k \check{\nu}_k) - 2\lambda_k(\alpha_k \check{\mu}_k + \beta_k \mu_k)}{\xi} \right) \check{e}_k \\ e_{-1} \cdot e_0 &= \gamma e_0 \\ e_{-1} \cdot e_k &= \left(\frac{\gamma(\mu_k \nu_k - \check{\mu}_k \check{\nu}_k) - 2\lambda_k(\alpha_k \mu_k - \beta_k \check{\mu}_k)}{2\lambda_k \xi} \right) e_0 + \gamma e_k \\ e_{-1} \cdot \check{e}_k &= \left(\frac{\gamma(\check{\mu}_k \nu_k + \mu_k \check{\nu}_k) - 2\lambda_k(\alpha_k \check{\mu}_k + \beta_k \mu_k)}{2\lambda_k \xi} \right) e_0 + \gamma \check{e}_k \\ e_k \cdot e_k &= \check{e}_k \cdot \check{e}_k = -\frac{\gamma}{2\lambda_k} e_0. \end{aligned}$$

Hence, we will consider two different situations.

$\gamma \neq 0$. By taking

$$\nu_k = \frac{2\alpha_k\lambda_k}{\gamma}, \check{\nu}_k = \frac{2\beta_k\lambda_k}{\gamma} \text{ and } \nu = -\sum_{i=1}^n \frac{\gamma(\nu_i^2 + \check{\nu}_i^2)}{2\gamma\lambda_i} - \frac{\mu}{\gamma},$$

we have a multiplication table from part (A) in our statement. It is easy to see that, under the action of a positive automorphism of the Lie algebra $(\mathfrak{L}_\lambda, [\cdot, \cdot])$, given in Lemma 30, all $\cdot$ -multiplications give non-isomorphic transposed Poisson structures. Under the action of a negative automorphism given in Lemma 30, we have the isomorphism between structures corresponding to γ and $-\gamma$. Insofar as $\gamma \neq 0$, $(\mathfrak{L}_\lambda, \cdot)$ is non-nilpotent.

$\gamma = 0$. In this case, $(\mathfrak{L}_\lambda, \cdot)$ is a nilpotent algebra. Note that,

- If $\{\alpha_k\}_{1 \leq k \leq n} = 0$, then transposed Poisson structures corresponding to $\{\beta_1, \dots, \beta_j, \dots, \beta_n\}$ and $\{\beta_1, \dots, -\beta_j, \dots, \beta_n\}$ are isomorphic.
- If $(\alpha_k, \beta_k) \neq 0$, then after an action of a suitable automorphism, we can suppose that $\alpha_k \neq 0$ and $\beta_k \neq 0$. In the case $(\alpha_k, \beta_k) = 0$ after each automorphism action, we will take the same situation $(\alpha_k, \beta_k) = 0$. Let t be the maximal number such that $(\alpha_1, \beta_1) = \dots = (\alpha_{t-1}, \beta_{t-1}) = 0$.

We should consider two subcases.

$\mu \neq 0$. By taking

$$\mu = \xi, \check{\mu}_t = \alpha_t \text{ and } \mu_k = \frac{\beta_k \check{\mu}_k}{\alpha_k} \text{ for all } k \text{ such that } (\alpha_k, \beta_k) \neq 0.$$

Hence, $(\mathfrak{L}_\lambda, \cdot)$ has a multiplication table from the part (B.a) of our statement.

$\mu = 0$. By taking

$$\check{\mu}_t = \alpha_t \text{ and } \mu_k = \frac{\beta_k \check{\mu}_k}{\alpha_k} \text{ for all } k \text{ such that } (\alpha_k, \beta_k) \neq 0.$$

Hence, $(\mathfrak{L}_\lambda, \cdot)$ has a multiplication table from the part (B.b) of our statement.

□

3. OPEN QUESTIONS

Question 1 (Pasha Zusmanovich). *Formulate the notion of "transposed" for any variety of algebras/operads with two n -ary operations. Could this be done in either operadic or in categorical theoretical language? This general notion should explain the appearance of "scaling by 2" in the definition of transposed Poisson algebras. What would be the transposed analogs of dendriform, post-Lie, Novikov-Poisson, etc.? What would be self-transposed algebras/operads?*

Thanks to [20, Theorem 8], we know that if a Lie algebra does not admit non-trivial $\frac{1}{2}$ -derivations, then it does not admit non-trivial $\frac{1}{2}$ -biderivations and transposed Poisson structures. All algebras which admit non-trivial $\frac{1}{2}$ -derivations (from the present paper and [20, 65]) also admit non-trivial $\frac{1}{2}$ -biderivations and non-trivial transposed Poisson structures. The last results are motivating the following question.

Question 2. *Is there a Lie algebra that admits non-trivial $\frac{1}{2}$ -derivations and does not admit non-trivial $\frac{1}{2}$ -biderivations? Is there a Lie algebra that admits non-trivial $\frac{1}{2}$ -derivations and $\frac{1}{2}$ -biderivations, and does not admit non-trivial transposed Poisson algebra structures?*

One of the first examples of non-trivial transposed Poisson algebras was given for the Witt algebra [20, Theorem 20]. On the other hand, the central extension of the Witt algebra is the Virasoro algebra and it has no non-trivial $\frac{1}{2}$ -derivations, and $\frac{1}{2}$ -biderivations and transposed Poisson algebra structures [20, Theorem 27]. The last results are motivating the following question.

Question 3. *Is there a Lie algebra which does not admit non-trivial transposed Poisson algebra structures ($\frac{1}{2}$ -derivations, $\frac{1}{2}$ -biderivations), but its central extension admits them?*

There are no non-trivial transposed Poisson algebra structures defined on a complex semisimple finite-dimensional Lie algebra [20, Corollary 9]. On the other hand, the Schrödinger algebra, which is the semidirect product of a simple and a nilpotent algebra, also does not have non-trivial transposed Poisson algebra structures. Theorem 14 states that each finite-dimensional nilpotent Lie algebra has a non-trivial transposed Poisson structure ($\frac{1}{2}$ -derivations, $\frac{1}{2}$ -biderivations). The last results are motivating the following question.

Question 4. *Is there a solvable (non-nilpotent) Lie algebra which does not admit non-trivial transposed Poisson algebra structures ($\frac{1}{2}$ -derivations, $\frac{1}{2}$ -biderivations)?*

The description of $\frac{1}{2}$ -derivations gives a good criterion for detection of Lie algebras that do not admit non-trivial transposed Poisson structures. Another criterion can be given by using Hom-Lie structures on Lie algebras (Proposition 19). However, the last criterion is not sufficient: the simple 3-dimensional algebra $\mathfrak{sl}_2$ has non-trivial Hom-Lie structures [61], but it does not admit non-trivial transposed Poisson structures [20, Corollary 9]. Filippov proved that each δ -derivation ($\delta \neq 0, 1$) gives a non-trivial Hom-Lie algebra structure [21, Theorem 1]. The last results are motivating the following questions.

Question 5. *Describe Lie algebras such that each Hom-Lie algebra structure gives a $\frac{1}{2}$ -derivation. Which types of Hom-Lie structures can guarantee the existence of non-trivial transposed Poisson structures on a Lie algebra?*

Let us denote the associative algebra generated by $\frac{1}{2}$ -derivations of $\mathfrak{L}$ as $\Delta_{\frac{1}{2}}(\mathfrak{L})$. The Witt algebra W has the following interesting property: $\mathfrak{Der}(\Delta_{\frac{1}{2}}(W)) \cong W$ [20, Theorem 19]. The last results are motivating the following question.

Question 6. *Is there a Lie algebra $\mathfrak{L}$ ($\not\cong W$) such that $\mathfrak{Der}(\Delta_{\frac{1}{2}}(\mathfrak{L})) \cong \mathfrak{L}$?*

Thanks to [20, Corollary 9], there are no non-trivial transposed Poisson algebra structures defined on a complex semisimple finite-dimensional Lie algebra; Theorem 6 establishes that there are no non-trivial transposed Poisson algebra structures defined on a complex semisimple associative commutative algebra. On the other hand, there are simple Lie algebras (the Witt algebra W and the Cartan algebra $W(1)$) that admit non-trivial transposed Poisson algebra structures, but in these cases the associative part is not simple (Theorem 24). The last results are motivating the following question.

Question 7. *Is there a (not necessarily commutative) transposed Poisson algebra $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ such that its Lie and associative parts are simple (semisimple, prime, semiprime) algebras?*

Combing results from Theorem 6 and [18, Proposition 28], we have that the Kantor product of a unital transposed Poisson algebra gives a new transposed Poisson algebra. Furthermore, the Kantor product of transposed Poisson algebras constructed on the Witt algebra gives new transposed Poisson algebras [18, Proposition 27]. The last results are motivating the following question.

Question 8. *Is there a transposed Poisson algebra such that the algebra constructed by the Kantor product of multiplications (see [18]) is not a transposed Poisson algebra?*

The Kantor double gives a way for a construction of Jordan superalgebras from Poisson algebras and from algebras of Jordan brackets [26]. As we can see from Corollary 9, each unital transposed Poisson algebra, by the Kantor double process, gives a Jordan superalgebra. The last results are motivating the following question.

Question 9. *Describe and study the class of superalgebras obtained by the Kantor double process from (not necessarily unital) transposed Poisson algebras.*

The celebrated Amitsur–Levitsky theorem states that the algebra Mat_k of $k \times k$ matrices over a commutative ring R satisfies the identity $s_{2k} = 0$. Furthermore, every associative PI algebra satisfies $(s_k)^l = 0$ by Amitsur’s theorem. Farkas defined customary Poisson polynomials and proved that every Poisson PI algebra satisfies some customary identity [17]. Recently, the Farkas’ theorem was established for generic Poisson algebras in [32], for generalized Poisson algebras in [28] and unital transposed Poisson algebras (Theorem 11). The last results are motivating the following question.

Question 10. *Is there a special type of polynomial identities for (not necessarily unital) transposed Poisson algebras satisfying a polynomial identity?*

In 1972, Regev proved that the tensor product of two associative PI algebras is a associative PI algebra. Recently, a similar result was obtained for Poisson algebras [49]. The last results are motivating the following question.

Question 11. *Is a tensor product of two transposed Poisson PI algebras a transposed Poisson PI algebra?*

It is known that the tensor product of two Poisson algebras gives a Poisson algebra [49] and also the tensor product of two transposed Poisson algebras gives a transposed Poisson algebra [4]. On the other hand, in the case of generalized Poisson algebras, their tensor product does not necessarily give a new generalized Poisson algebra [70]. The last results are motivating the following question.

Question 12 (Pasha Zusmanovich). *Find identities for algebras formed by the tensor product of a transposed Poisson algebra and a Poisson algebra.*

The notion of an admissible Poisson algebra (i.e., a binary algebra such that the under commutator and the anticommutator products give a Poisson algebra) was introduced and studied in [16]. The last results are motivating the following question.

Question 13. *Define and study the variety of admissible transposed Poisson algebras.*

In 1950, Hall constructed a basis for free Lie algebras. Later on: Shestakov constructed a basis for free Poisson algebras; Shestakov and Zhukavets constructed a basis for free Poisson–Malcev superalgebra with one generator; Kaygorodov, Shestakov and Umirbaev constructed a basis for free generic Poisson algebras in [30]; and Kaygorodov constructed a basis for free unital generalized Poisson algebras in [28]. The last results are motivating the following question.

Question 14. *Construct a basis for free transposed Poisson algebras.*

In 1930, Magnus proved one of the most important theorems of the combinatorial group theory: Let $G = \langle x_1, x_2, \dots, x_n \mid r = 1 \rangle$ be a group defined by a single cyclically reduced relator r . If x_n appears in r , then the subgroup of G generated by $x_1, \dots, x_{n-1}$ is a free group, freely generated by $x_1, \dots, x_{n-1}$. He called it the Freiheitssatz and used it to give several applications, the decidability of the word problem for groups with a single defining relation among them. The Freiheitssatz was studied in various varieties of groups and algebras. Thus, Shirshov established it for Lie algebras; Romanovskii researched it for solvable and nilpotent groups; Makar-Limanov proved it for associative algebras over a field of characteristic zero. Recently it was confirmed for Novikov and right-symmetric algebras, as well as for Poisson algebras [43] and generic Poisson algebras [32]. The last results are motivating the following question.

Question 15. *Is the Freiheitssatz true for transposed Poisson algebras?*

In 1942, Jung proved that all automorphisms of the polynomial algebra in two variables are tame. Later, in 1970, Makar-Limanov established the same result for the free associative algebra in two variables. Cohn proved that the automorphisms of a free Lie algebra with a finite set of generators are tame. Shestakov and Umirbaev proved that polynomial algebras and free associative algebras in three variables in the case of characteristic zero have wild automorphisms [53, 57]. At the same time, Abdykhalykov, Mikhalev and Umirbaev found wild automorphisms in two generated free Leibniz algebras. It was recently proved that free Poisson and generic Poisson algebras in two variables do not have wild automorphisms [30, 41]. The last results are motivating the following question.

Question 16. *Are the automorphisms of a two-generated free transposed Poisson algebra tame?*

Makar-Limanov and Umirbaev [42] proved an analog of the Bergman Centralizer Theorem: the centralizer of every non-constant element in a free Poisson algebra is a polynomial algebra in one variable. The last results are motivating the following question.

Question 17 (Zerui Zhang). *Describe the centralizer of a non-constant element in a free transposed Poisson algebra.*

Thanks to Corollary 9, we know that each unital transposed Poisson algebra is an algebra of Jordan brackets. Martínez, Shestakov and Zelmanov have shown that all Jordan brackets are embeddable into Poisson brackets [46]. Hence, all unital transposed Poisson brackets are embeddable into Poisson brackets. The last results are motivating the following question.

Question 18. *Are transposed Poisson brackets embeddable into Poisson brackets?*

Thanks to a famous relationship between Poisson algebras and Jordan superalgebras, also known as Kantor double, the description of simple Jordan superalgebras describes simple Poisson algebras. Also, recently, simple generalized Poisson algebras and simple generalized Gerstenhaber algebras (odd generalized Poisson superalgebras) have been described [11, 12]. The last results are motivating the following question.

Question 19. *Classify simple transposed Poisson algebras.*

The study of noncommutative Poisson algebras, which were introduced in [63], and Theorem 6 are motivating the following question.

Question 20. *Define and study noncommutative transposed Poisson algebras.*

One of the important questions in the study of Poisson and transposed Poisson algebras is a description of (transposed) Poisson algebra structures on a certain Lie algebra or on a certain associative algebra. The most useful way for a description of Poisson structures on a certain associative algebra is given by the study of derivations and biderivations of this algebra. On the other hand, a description of transposed Poisson structures on a certain Lie algebra may be given by the study of $\frac{1}{2}$ -derivations and $\frac{1}{2}$ -biderivations of this algebra. Recently, a constructive method for describing Poisson algebra structures on a certain Lie algebra has been given in a study of Leibniz bialgebras and symmetric Leibniz algebras [1]. The last results and Theorem 6 are motivating the following question.

Question 21. *Give a constructive method for describing transposed Poisson algebras on a certain non-unital associative (not necessarily commutative) algebra.*

It is known that the Poisson bracket admits an extension to the field of fractions [25, 30, 40, 44]. As we proved in Theorem 12, a unital transposed Poisson bracket admits an extension to the field of fractions. The last results are motivating the following question.

Question 22. *Does the bracket in Theorem 12 give an extension of a transposed Poisson algebra to the field of fractions in the non-unital case?*

The universal multiplicative enveloping algebra of free Poisson and free generic Poisson fields are studied in [30, 44]. In particular, in these papers it was proved that the universal multiplicative enveloping algebra is a free ideal ring. The last results are motivating the following question.

Question 23. *Is the universal multiplicative enveloping algebra of a free transposed Poisson field a free ideal ring?*

It is known that each Novikov-Poisson algebra under commutator product on non-associative multiplication gives a transposed Poisson algebra [4]. Let us say that a transposed Poisson algebra is special if it can be embedded into a Novikov-Poisson algebra relative to the commutator bracket. Similarly, let us say that a transposed Poisson algebra is D-special (from “differentially”) if it embeds into a commutative algebra with a derivation relative to the bracket $[x, y] = \mathfrak{D}(x)y - x\mathfrak{D}(y)$. Obviously, every D-special transposed Poisson algebra is a special one. It is known, that the class of special Jordan algebras (i.e. embedded into associative algebras relative to the multiplication $x \circ y = xy + yx$)

is a quasivariety, but it is not a variety of algebras [56]. On the other hand, the class of all special Gelfand-Dorfman algebras (i.e. embedded into Poisson algebras with derivation relative to the multiplication $x \circ y = xd(y)$) is closed with respect to homomorphisms and thus forms a variety [33]. The last results are motivating the following question.

Question 24 (Pavel Kolesnikov). *Whether the class of special (resp., D -special) transposed Poisson algebras forms a variety or a quasivariety?*

Let us call by special identities all identities that hold on all special transposed Poisson algebras but do not hold on the class of all transposed Poisson algebras. The study of special and non-special identities is an popular topic in non-associative algebras (see, for example, Jordan algebra case in [23], Gelfand-Dorfman algebra case in [32], dialgebras [35], Jordan trialgebra and post-Jordan algebra cases in [3], etc.). The last results are motivating the following question.

Question 25 (Pavel Kolesnikov). *Find the identities separating the variety generated by special (resp., D -special) transposed Poisson algebras in the variety of all transposed Poisson algebras. Find the identities separating the variety generated by D -special transposed Poisson algebras in the variety generated by special transposed Poisson algebras.*

Also known as, each two-generated Jordan algebra is special [55]; each two-generated Jordan dialgebra is special [59]; each 2-dimensional Gelfand-Dorfman algebra is special [33]. The last results are motivating the following question.

Question 26. *Are one- and two-generated transposed Poisson algebras special (resp., D -special)? Are 2- and 3-dimensional transposed Poisson algebras special (resp., D -special)?*

On the geometric point of view, the variety of n -dimensional algebras defined by a family of polynomial identities gives a subvariety in $\mathbb{C}^{n^3}$. The geometric classification of a variety of algebras is a description of his geometric variety: defining irreducible components, their dimensions and rigid algebras (see, [2, 19, 31] and references in therein). The variety of transposed Poisson algebras gives a subvariety in $\mathbb{C}^{2n^3}$. The last results are motivating the following question.

Question 27. *Give the geometric classification of low-dimensional transposed Poisson algebras.*

Neretin in his paper [50] considered n -dimensional associative, commutative and Lie algebras and gave the upper bound for dimensions of their geometric varieties. The last results are motivating the following question.

Question 28. *Give the upper bound for dimensions of geometric varieties of n -dimensional transposed Poisson algebras.*

The “Koszul” property of an (quadratic) operad is quite important. The explicit definition needs more notions on operads [22, 38]. If an operad P is Koszul, one can gives the ∞ -structure of such an operad, P_∞ -algebra. For example, Lie, associative, pre-Lie, and Poisson are Koszul and hence we have L_∞ -, A_∞ -, PL_∞ -, P_∞ -algebras.

Question 29 (Li Guo). *Whether the operad of the transposed Poisson is Koszul? If so, there might an TPA_∞ -algebra structure.*

The Poisson operad is isomorphic to the Manin white product of commutative associative and Lie, which motivating the following question.

Question 30 (Pasha Zusmanovich). *Decompose the transposed Poisson operad in the Manin product.*

Some recently works about cohomology theory of Poisson algebras [6, 52] are motivating the following question.

Question 31. *Describe the cohomology theory of transposed Poisson algebras.*

Poisson algebras are the semi-limit of quantization deformation of commutative associative algebras into associative algebras. F -manifold algebras are the semi-limit of quantization deformation of commutative associative algebras (or commutative pre-Lie algebras) into pre-Lie algebras. The last results are motivating the following question.

Question 32 (Chengming Bai). *Into which algebras, the transposed Poisson algebras are the semi-limit of quantization deformation of commutative algebras.*

The study of commutative and noncommutative Poisson bialgebras [37, 51] are motivating the following question.

Question 33 (Chengming Bai). *Define and study transposed Poisson bialgebras.*

The study of double Poisson algebras [58] are motivating the following question.

Question 34. *Define and study double transposed Poisson algebras.*

Bai, Bai, Guo and Wu proved that each transposed Poisson algebra $(\mathfrak{L}, \cdot, [\cdot, \cdot])$ gives a transposed Poisson 3-Lie algebra by defining the following multiplication:

$$[x, y, z] = \mathfrak{D}(x)[y, z] - \mathfrak{D}(y)[x, z] + \mathfrak{D}(z)[x, y].$$

The last results are motivating the following question.

Question 35 (Chengming Bai, Ruipu Bai, Li Guo & Yong Wu). *Let $n \geq 2$ be an integer. Let $(\mathfrak{L}, \cdot, [\cdot, \dots, \cdot])$ be a transposed Poisson n -Lie algebra and let $\mathfrak{D}$ be a derivation of $(\mathfrak{L}, \cdot)$ and $(\mathfrak{L}, [\cdot, \dots, \cdot])$. Define an $(n + 1)$ -ary operation*

$$[[x_1, \dots, x_{n+1}]] := \sum_{i=1}^{n+1} (-1)^{i+1} \mathfrak{D}(x_i)[x_1, \dots, \hat{x}_i, \dots, x_{n+1}],$$

where $\hat{x}_i$ means that the i -th entry is omitted. *Is $(\mathfrak{L}, \cdot, [[\cdot, \dots, \cdot]])$ a transposed Poisson $(n + 1)$ -Lie algebra?*

Cantarini and Kac classified all complex linearly compact (generalized) Poisson n -Lie algebras in [13], which motivates the following question.

Question 36. *Give examples and classify simple transposed Poisson n -Lie algebras.*

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